WellBridge: Project Blueprint & Implementation Guide

1. Core Concept & Strategy

- **The Problem:** Active individuals skip workouts and break habits when they arrive at their usual facility only to find it overcrowded, lacking immediate visibility into alternative nearby sports clubs or gyms.
- **The Solution:** A unified wellness access platform that shows live crowd levels and allows users to pay once on the app to freely check into any less crowded partner gym, boxing ring, or sports club nearby.
- **The Value Proposition (One-Liner):** My app helps busy gym-goers to never skip a workout by unlocking nearby open spots with non-expiring digital credits.
- **Revenue Model:** The platform earns a percentage-based transaction fee per check-in from the user's digital wallet, with secondary growth driven by selling bulk non-expiring wellness credits to corporate HR programs.

2. Must-Have Features

1. **Non-Expiring Digital Wallet:** Pay-per-visit access using loaded credits that roll over indefinitely.
2. **Live Capacity Map:** Real-time visibility into current crowd levels at nearby facilities.
3. **Universal QR Check-in:** A single dynamic scan code for instant front-desk entry at all partner locations.
4. **AI Alternative Routing:** Automatically predicts peak hours and suggests the nearest open spot if a primary location is full.
5. **Partner Revenue Dashboard:** A dedicated portal for facility owners to track live incoming check-ins and revenue share.

3. Recommended Modern Tech Stack

To build a highly scalable, offline-capable, and fast cross-platform application, utilize this full-stack architecture:

- **Frontend Mobile/Web:** React 19 with TypeScript 5 and Vite 6, packaged for mobile using Capacitor 8.
- **Styling:** Tailwind CSS v4 for rapid, responsive UI development.
- **Backend Services:** Python FastAPI for high-performance, asynchronous API routing.
- **Database:** PostgreSQL for robust transactional data (wallet balances, facility check-ins).
- **Mapping & Location:** OpenStreetMap for plotting local facilities and generating the live capacity map without excessive API costs.
- **Hosting & Deployment:** Google Cloud Run for the FastAPI backend and Firebase Hosting for the frontend web application.

- **AI Integration:** Gemini API (via Google AI Studio) for routing algorithms and predicting facility capacity trends based on historical check-in data.

4. Google AI Studio Setup & Prompting

Based on your current workspace, here is how to configure Google AI Studio to generate the frontend components for WellBridge.

Recommended AI Studio Settings

Update your Advanced Settings to match your exact tech stack:

- **Model:** Default (Gemini 3.8 Flash)
- **Framework:** React
- **System Instructions:** Replace the existing NGO-focused instructions with the following to align with your new stack:
  "You are an expert Frontend Web Developer and UI/UX Architect specializing in high-conversion fitness and wellness platforms. Your objective is to generate clean, modular, and production-ready code utilizing React 19, TypeScript, and Tailwind CSS v4. Ensure the architecture is highly scalable, fast, and optimized for mobile-first views via Capacitor."

The Generation Prompt

Copy and paste this prompt into Google AI Studio to generate your first critical interface:

Prompt:

"Build the 'Live Capacity & Discovery Map' screen for WellBridge, a mobile-first fitness access app.

Requirements:

1. Create a React functional component using TypeScript and Tailwind CSS.
2. The top section should feature a clean greeting ("Hi, User!") and display their current "Digital Wallet Balance" in non-expiring credits.
3. The main body should be a placeholder for an interactive map (simulate with a visually appealing styled div), overlaying 3 nearby facility pins (e.g., a gym, a boxing club, and a calisthenics park).
4. Below the map, display a horizontal scrollable list of these 3 facilities as cards. Each card must show the facility name, distance, and a prominent 'Live Capacity' indicator (e.g., '60% Full - Good to Go' or '95% Full - Crowded').
5. Include a bottom navigation bar with icons for 'Home/Map', 'Wallet', and 'Profile'.

Keep the design modern, high-contrast, and suitable for someone actively on the move."